```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
df = pd.read_csv("dataset.csv")
df.head()
```

	itching	skin_rash	nodal_skin_eruptions	continuous_sneezing	shivering	chills	joint_pain	s
0	1	1	1	0	0	0	0	
1	0	1	1	0	0	0	0	
2	1	0	1	0	0	0	0	
3	1	1	0	0	0	0	0	
4	1	1	1	0	0	0	0	

5 rows × 134 columns

```
df.columns = df.columns.str.strip()
df.columns = df.columns.str.replace(" ", "_")
df.columns = df.columns.str.replace("(", "", regex=False)
df.columns = df.columns.str.replace(")", "", regex=False)

df = df.drop_duplicates()

df.head()
```

	itching	skin_rash	nodal_skin_eruptions	continuous_sneezing	shivering	chills	joint_pain	s
0	1	1	1	0	0	0	0	
1	0	1	1	0	0	0	0	
2	1	0	1	0	0	0	0	
3	1	1	0	0	0	0	0	
4	1	1	1	0	0	0	0	

5 rows × 134 columns

```
X = df.drop("prognosis", axis=1)
y = df["prognosis"]

X = X.apply(pd.to_numeric, errors='coerce').fillna(0)

print("Shape of X:", X.shape)
print("Shape of y:", y.shape)
```

```
Shape of X: (304, 133)
Shape of y: (304,)
```

```
np.random.seed(42)

noise_factor = 0.04

X_noisy = X.copy()

for col in X.columns:
    mask = np.random.rand(len(X)) < noise_factor
    X_noisy.loc[mask, col] = 1 - X_noisy.loc[mask, col]

X_noisy.head()
```

	itching	skin_rash	nodal_skin_eruptions	continuous_sneezing	shivering	chills	joint_pain	s
0	1	1	1	0	0	0	0	
1	0	1	1	0	0	0	0	
2	1	0	0	0	0	0	0	
3	1	1	0	0	0	0	0	
4	1	1	1	0	0	0	0	

5 rows × 133 columns

```
X_train, X_test, y_train, y_test = train_test_split(
    X_noisy,
    y,
    test_size=0.3,
    random_state=42,
    stratify=y
)

print("Train size:", X_train.shape)
print("Test size:", X_test.shape)
```

```
Train size: (212, 133)
Test size: (92, 133)
```

```
model = RandomForestClassifier(
    n_estimators=120,
    max_depth=12,
    min_samples_split=6,
    min_samples_leaf=3,
    random_state=42
)

model.fit(X_train, y_train)
```

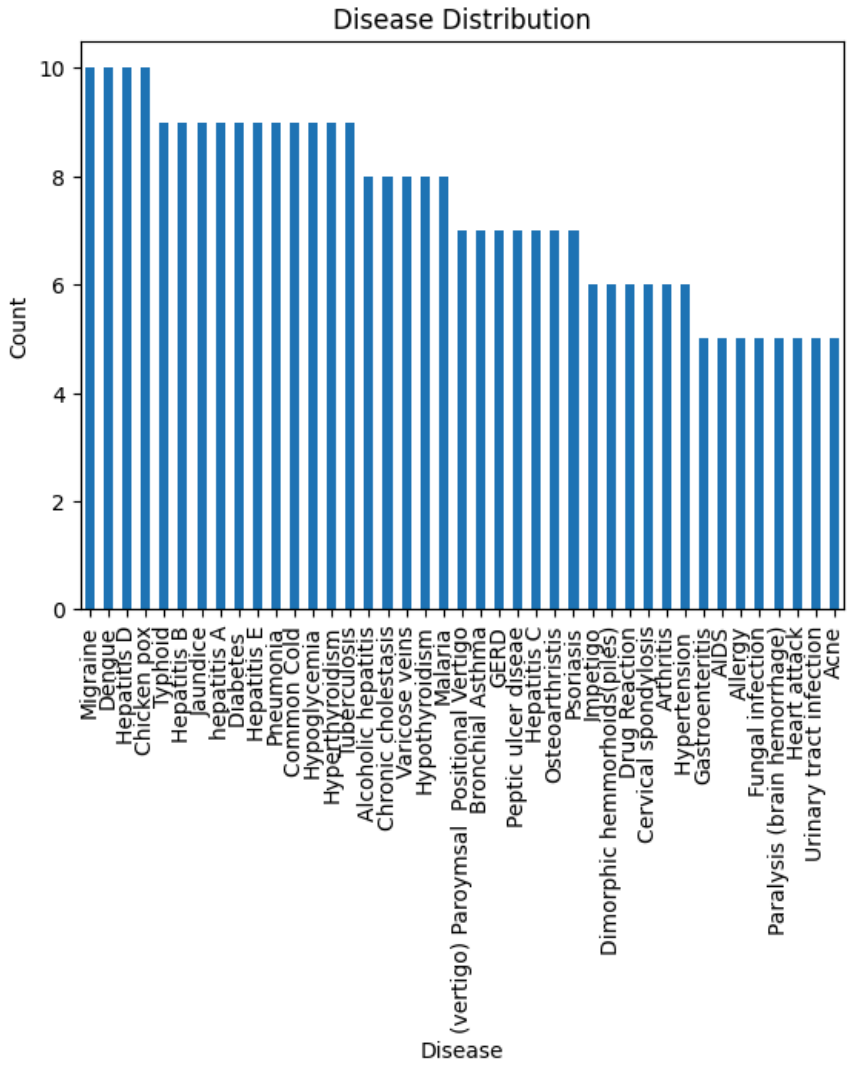
```
▾ RandomForestClassifier
RandomForestClassifier(max_depth=12, min_samples_leaf=3, min_samples_split=6,
                       n_estimators=120, random_state=42)
```

```
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nClassification Report:\n")
print(classification_report(y_test, y_pred, zero_division=0))

print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, y_pred))
```

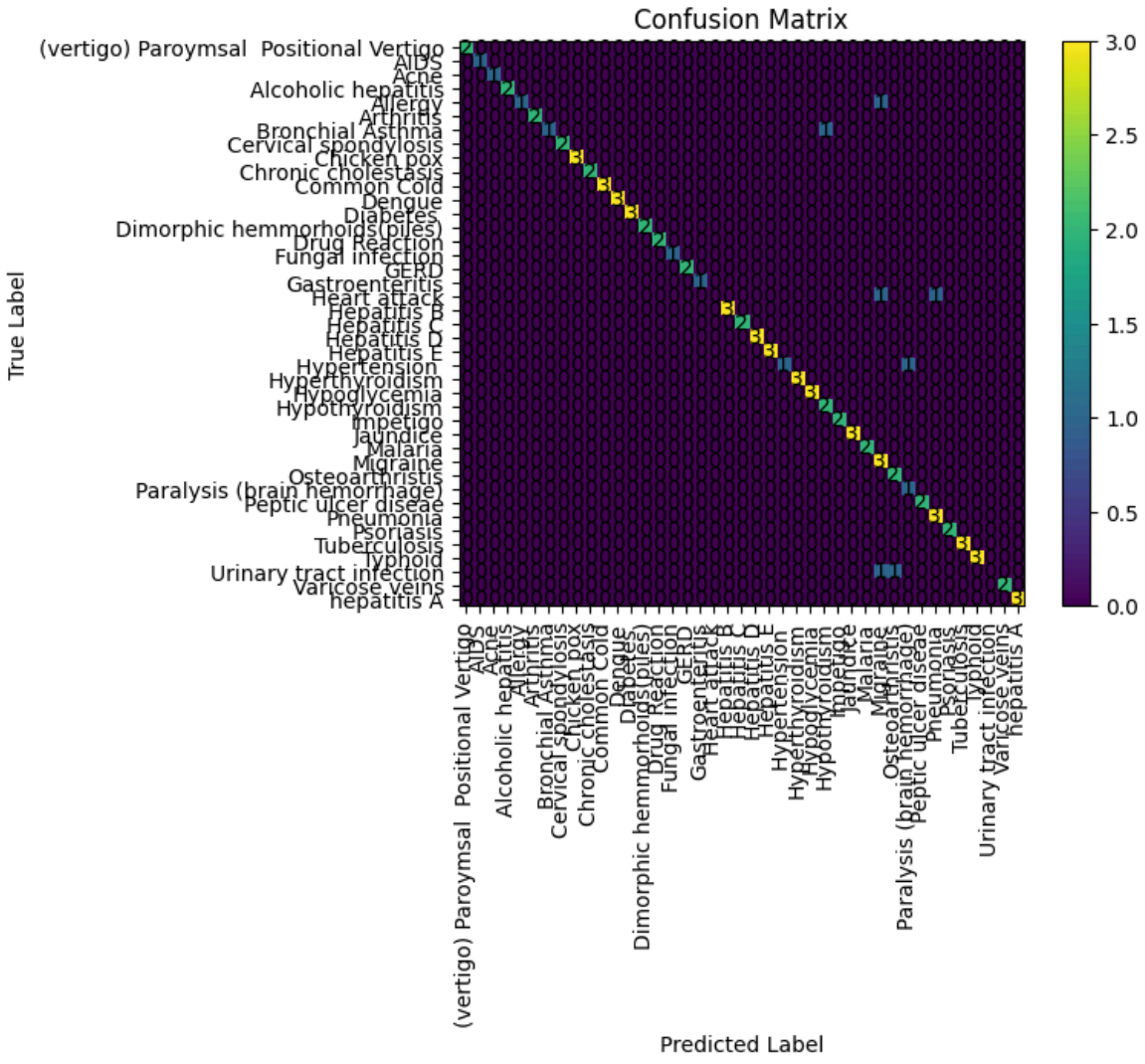
```
Accuracy: 0.9239130434782609

Classification Report:

                                          precision    recall  f1-score   support
```

```
(vertigo) Paroymsal  Positional Vertigo       1.00      1.00      1.00         2
                                   AIDS       1.00      1.00      1.00         1
                                   Acne       1.00      1.00      1.00         1
                     Alcoholic hepatitis       1.00      1.00      1.00         2
                                 Allergy       1.00      0.50      0.67         2
                               Arthritis       1.00      1.00      1.00         2
                        Bronchial Asthma       1.00      0.50      0.67         2
                    Cervical spondylosis       1.00      1.00      1.00         2
                             Chicken pox       1.00      1.00      1.00         3
                     Chronic cholestasis       1.00      1.00      1.00         2
                             Common Cold       1.00      1.00      1.00         3
                                  Dengue       1.00      1.00      1.00         3
                               Diabetes        1.00      1.00      1.00         3
            Dimorphic hemmorhoids(piles)       1.00      1.00      1.00         2
                          Drug Reaction       1.00      1.00      1.00         2
                        Fungal infection       1.00      1.00      1.00         1
                                    GERD       1.00      1.00      1.00         2
                       Gastroenteritis       1.00      1.00      1.00         1
                          Heart attack       0.00      0.00      0.00         2
                           Hepatitis B       1.00      1.00      1.00         3
                           Hepatitis C       1.00      1.00      1.00         2
                           Hepatitis D       1.00      1.00      1.00         3
                           Hepatitis E       1.00      1.00      1.00         3
                         Hypertension        1.00      0.50      0.67         2
                       Hyperthyroidism       1.00      1.00      1.00         3
                          Hypoglycemia       1.00      1.00      1.00         3
                        Hypothyroidism       0.67      1.00      0.80         2
                               Impetigo       1.00      1.00      1.00         2
                               Jaundice       1.00      1.00      1.00         3
                                Malaria       1.00      1.00      1.00         2
                               Migraine       0.50      1.00      0.67         3
                       Osteoarthristis       0.67      1.00      0.80         2
            Paralysis (brain hemorrhage)       0.50      1.00      0.67         1
                     Peptic ulcer diseae       1.00      1.00      1.00         2
                               Pneumonia       0.75      1.00      0.86         3
                               Psoriasis       1.00      1.00      1.00         2
                          Tuberculosis       1.00      1.00      1.00         3
                                Typhoid       1.00      1.00      1.00         3
                 Urinary tract infection       0.00      0.00      0.00         2
                        Varicose veins       1.00      1.00      1.00         2
                           hepatitis A       1.00      1.00      1.00         3

                               accuracy                           0.92        92
                              macro avg       0.90      0.91      0.90        92
                           weighted avg       0.91      0.92      0.91        92


Confusion Matrix:

[[2 0 0 ... 0 0 0]
 [0 1 0 ... 0 0 0]
```

```
import matplotlib.pyplot as plt

plt.figure()
y.value_counts().plot(kind='bar')

plt.title("Disease Distribution")
plt.xlabel("Disease")
plt.ylabel("Count")
plt.xticks(rotation=90)

plt.show()
```




```
import matplotlib.pyplot as plt
import numpy as np

cm = confusion_matrix(y_test, y_pred)

plt.figure()
plt.imshow(cm)

# Add labels (actual disease names)
classes = np.unique(y_test)

plt.xticks(ticks=np.arange(len(classes)), labels=classes, rotation=90)
plt.yticks(ticks=np.arange(len(classes)), labels=classes)

# Add numbers inside boxes
for i in range(cm.shape[0]):
    for j in range(cm.shape[1]):
        plt.text(j, i, cm[i, j], ha="center", va="center")

plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")

plt.colorbar()
plt.show()
```




```
importances = pd.Series(model.feature_importances_, index=X.columns)

print("Top Symptoms:\n")
print(importances.sort_values(ascending=False).head(10))
```

```
Top Symptoms:

muscle_pain         0.027629
fatigue             0.026485
vomiting            0.023411
joint_pain          0.022902
abdominal_pain      0.022410
nausea              0.021765
loss_of_appetite    0.021337
headache            0.018948
mild_fever          0.018742
malaise             0.018410
dtype: float64
```

```
def predict_disease(symptoms_list):

    input_data = pd.DataFrame(
        np.zeros((1, len(X.columns))),
        columns=X.columns
    )

    for symptom in symptoms_list:
        symptom = symptom.strip().replace(" ", "_")
        if symptom in input_data.columns:
            input_data[symptom] = 1

    prediction = model.predict(input_data)
```